

Report on the Radiomics Classification Project

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1 Initial Dataset Review and Feature Analysis

This notebook was run on the **ADC_M1 dataset**.

1.1 Dataset Preprocessing

After cleaning, the ADC dataset contained:

- 114 ROIs;
- 48 unique patients;
- 145 numerical radiomics features;
- 50 benign and 64 malignant ROIs.

Some patients had more than one ROI. Therefore, patient ID was used as the grouping variable during cross-validation. All ROIs from one patient were kept in the same fold, so the same patient could not appear in both the training and test sets. This was necessary to prevent patient-level data leakage.

1.2 Data Quality Control

The main data-quality results were:

- total missing values: 12;
- features containing missing values: 6;
- infinite values: 0;
- constant features: 0;
- near-zero variance features: 0;
- duplicated rows: 0;
- duplicated ROI names: 0;
- duplicated patient-ROI pairs: 0.

The missing values were not filled before cross-validation. Median imputation was performed only using the training data of each fold.

The number of ROIs per patient ranged from 1 to 7, with a median of 2 and a mean of 2.38. Four patients had both benign and malignant ROIs:

ADC_P3, ADC_P50, ADC_P12, and ADC_P27.

1.3 Outlier Analysis

At least one outlier was found in 104 of the 145 features, which is about 72% of the feature set. The largest numbers of outliers were observed in:

- GLSZM_SmallZoneHighGreyLevelEmphasis: 44;
- GLRLM_ShortRunHighGreyLevelEmphasis: 41;
- GLRLM_LongRunsEmphasis: 33;
- IH_MaximumHistogramGradientGreyLevel_Intensity: 29;
- GLSZM_SmallZoneEmphasis: 29.

No global clipping, winsorisation, or sample removal was applied. This avoided changing the complete dataset before validation. However, robust scaling or fold-specific outlier handling should be tested later because some models may be sensitive to extreme values.

1.4 Class Distribution

The dataset contained 50 benign and 64 malignant ROIs. The majority-to-minority ratio was 1.28.

The exact binomial test gave:

$$p = 0.223.$$

Therefore, the class distribution was not significantly different from a balanced distribution. The imbalance was mild, so SMOTE was tested as an optional method rather than being applied automatically.

1.5 Feature Distribution and Statistical Tests

The normality analysis showed:

- 6 approximately normal features;
- 139 non-normal features.

For this reason, non-parametric methods such as the Mann–Whitney test and Spearman correlation were mainly used.

After false discovery rate correction, eight features showed a significant class difference.

Feature	Cohen's d
Morph_SurfaceToVolumeRatio_cm	0.566
Morph_Compacity	-0.621
Morph_Compactness1	-0.627
Morph_Compactness2	0.579
Morph_SphericalDisproportion	-0.616
Morph_Sphericity	0.579
GLSZM_LargeZoneEmphasis	-0.405
GLSZM_LowGrayLevelZoneEmphasis	0.496

Table 1: ADC features with significant class differences after FDR correction.

Most significant features were related to lesion geometry, while two GLSZM features described texture and zone-size patterns. This suggests that both shape and texture contained useful information for separating the two classes.

The morphological features were strongly correlated. Therefore, they should not be interpreted as independent biomarkers. They mainly describe related aspects of lesion geometry.

1.6 Unidentified Feature Column

I found an additional column named **FEATURE_RESULTS** in the **ADC_M1** dataset. I do not know its exact meaning clearly. It may have been added by mistake. This column should be checked before the final analysis.

1.7 Visual Review of the Dataset

1.7.1 Top FDR-Ranked Features

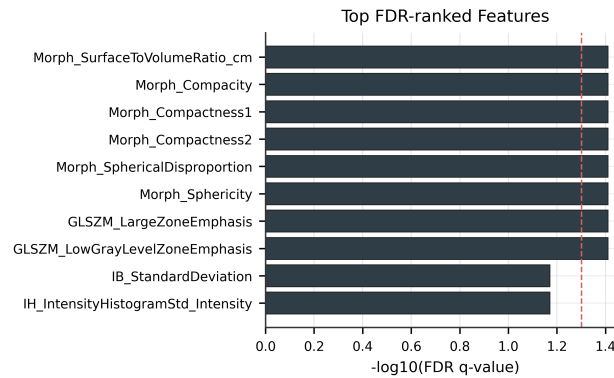


Figure 1: Top ADC radiomics features ranked by FDR-adjusted p-value.

The highest-ranked features were mainly morphological, with two GLSZM texture features also passing the FDR threshold. Therefore, lesion geometry and texture were the strongest individual signals in the ADC dataset.

1.7.2 Correlation among Important Features

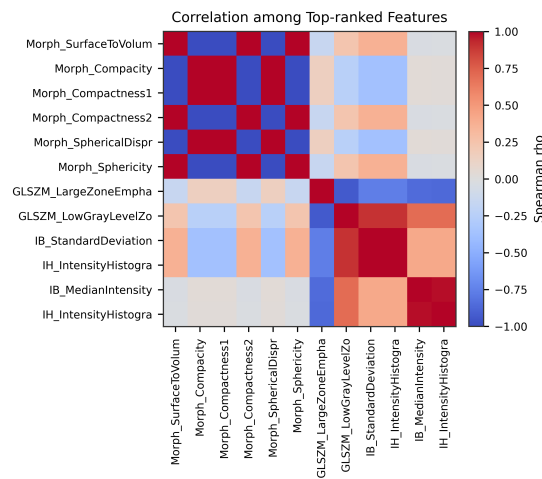


Figure 2: Spearman correlation among the highest-ranked ADC features.

Several important features were strongly correlated. This showed that some variables contained repeated information and supported the use of redundancy-aware feature-selection methods such as mRMR.

1.7.3 Feature Variance

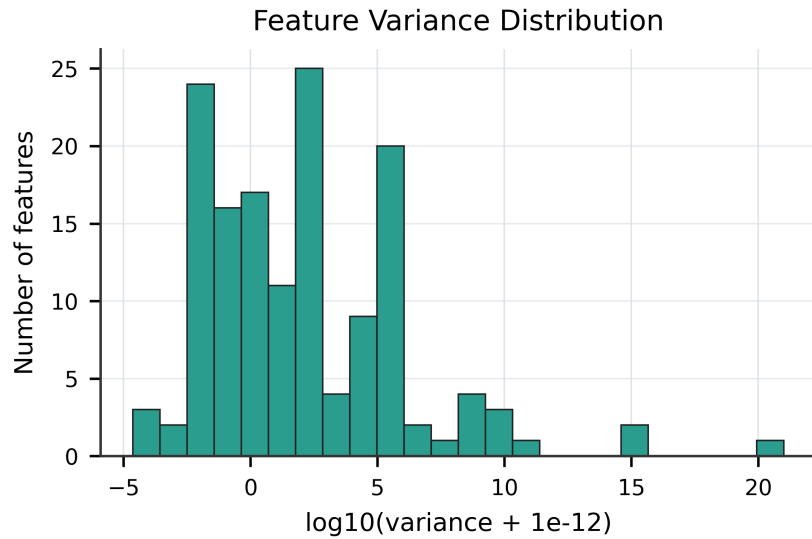


Figure 3: Distribution of ADC radiomics feature variance.

The features had very different numerical scales. No constant or near-zero variance features were found, but standardisation was necessary for scale-sensitive models such as Logistic Regression, SVM, and KNN.

1.7.4 Principal Component Analysis

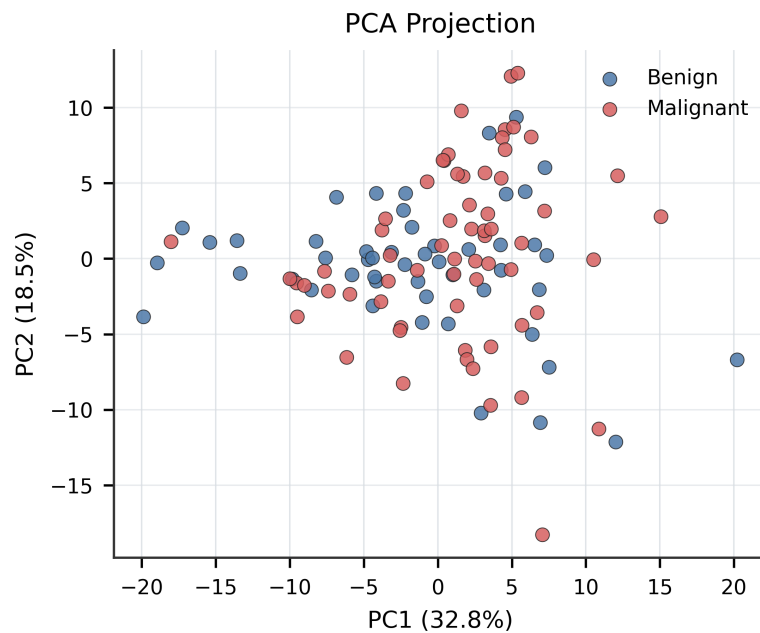


Figure 4: PCA projection of benign and malignant ADC ROIs.

The first two principal components explained approximately 51.2% of the total variance:

$$PC1 = 32.8\%, \quad PC2 = 18.5\%.$$

The benign and malignant samples still showed considerable overlap. Therefore, the classes were not naturally separated in a simple low-dimensional space, and supervised feature selection and classification were required.

2 Baseline, Model Definition, and Main Methods

2.1 Baseline Experiment

A fixed baseline experiment was first used to test the complete workflow. It used:

- mRMR followed by ElasticNet;
- a fixed number of 14 selected features;
- no SMOTE;
- five patient-aware outer folds;
- three patient-aware inner folds;
- Optuna for hyperparameter tuning.

The best baseline model was Logistic Regression:

$$MeanAUC = 0.707 \pm 0.078, \quad PooledAUC = 0.687.$$

The baseline showed moderate predictive performance, but the fold-to-fold variation was relatively high.

2.2 Feature-Selection Methods

Several feature-selection methods were implemented, including LASSO, RFE, a Boruta-like method, and custom mRMR. However, the main comparison focused on three pipelines.

1. ElasticNet followed by mRMR

ElasticNet first reduced the feature space, and mRMR then selected relevant features with lower redundancy. This was the strongest method for ADC. Without SMOTE, Logistic Regression achieved:

$$0.771 \pm 0.038.$$

2. Correlation filtering followed by mRMR

Highly correlated features were first reduced, and mRMR was then applied. This method gave intermediate but stable performance:

$$0.745 \pm 0.026$$

without SMOTE. It was weaker than ElasticNet followed by mRMR, but its low standard deviation showed good fold stability.

3. mRMR followed by ElasticNet

mRMR was applied first, followed by ElasticNet.

This order was the weakest main strategy for ADC. Its best result without SMOTE was:

$$0.711 \pm 0.075.$$

The comparison showed that the order of feature selection was important. Reversing the baseline order and applying ElasticNet before mRMR improved the mean AUC from 0.707 to 0.771 and reduced the standard deviation from 0.078 to 0.038.

2.3 Automatic Selection of Feature Number

The feature count was fixed at 14 only in the baseline. In the main experiments, the number of features was selected automatically from:

$$k \in \{4, 6, 8, 10, 12, 14, 16, 18\}.$$

For the best ElasticNet–mRMR pipeline, the selected numbers across the five outer folds were:

$$14, 8, 14, 18, 18.$$

The average was 14.4 features. Therefore, the final pipeline did not use one fixed feature count. The selected complexity changed according to the training patients in each fold.

2.4 Machine Learning Models

Ten classifiers were compared:

- Logistic Regression;
- linear SVM;
- RBF SVM;
- Random Forest;
- Extra Trees;
- Gradient Boosting;
- K-Nearest Neighbours;
- Naive Bayes;
- XGBoost;
- LightGBM.

The strongest results were obtained by simple linear models. Logistic Regression produced the highest mean AUC, while linear SVM produced a competitive result when SMOTE was used.

2.5 Effect of SMOTE

SMOTE was applied only to training data. It was never applied to validation or test samples.

Its effect depended on the classifier:

- with ElasticNet followed by mRMR, Logistic Regression decreased from 0.771 to 0.752;
- with the same feature-selection method, linear SVM increased from 0.735 to 0.764;
- the pooled AUC of linear SVM increased from 0.663 to 0.720;
- with correlation filtering followed by mRMR, SMOTE had almost no effect on Logistic Regression;
- with mRMR followed by ElasticNet, SMOTE slightly reduced Logistic Regression performance.

Therefore, SMOTE was useful for linear SVM but not for the best Logistic Regression model. It should be considered a model-specific option rather than a required preprocessing step.

2.6 Hyperparameter Optimisation

Optuna was used inside the inner patient-aware folds to tune the main parameters of each model.

2.7 Experimental Grid

The three main feature-selection methods were tested with and without SMOTE. Each setting was evaluated using all ten classifiers:

$$3 \times 2 \times 10 = 60$$

model configurations.

According to the primary metric, mean outer-fold AUC, the best configuration was:

ElasticNet + mRMR + Auto-k + Logistic Regression, without SMOTE.

Its main results were:

$$\text{MeanAUC} = 0.771 \pm 0.038, \quad \text{PooledAUC} = 0.710.$$

A closely competitive model used the same feature-selection method with SMOTE and linear SVM:

$$\text{MeanAUC} = 0.764 \pm 0.051, \quad \text{PooledAUC} = 0.720.$$

The Logistic Regression model had the highest mean AUC and lower variability, while the SVM model had a slightly higher pooled AUC and balanced accuracy.

2.8 Nested Cross-Validation

The complete pipeline was evaluated using patient-aware nested cross-validation. This design reduced data leakage and provided out-of-fold predictions for all samples.

3 Baseline Model Results

3.1 ROC Curves

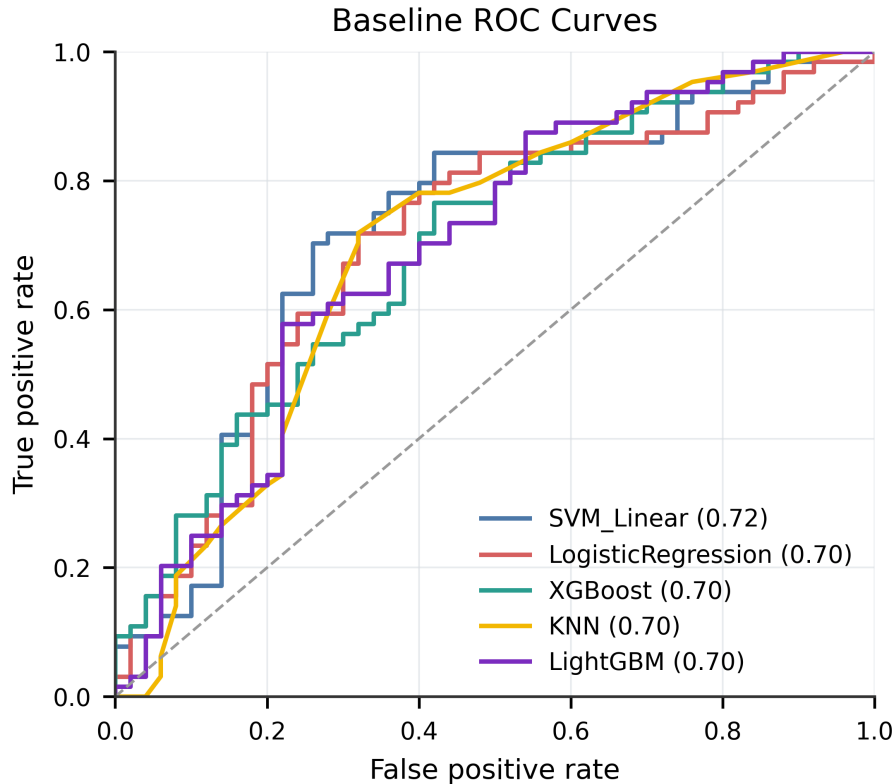


Figure 5: ROC curves of the strongest baseline models.

All displayed models performed better than random classification. Linear SVM had the highest ROC-AUC.

3.2 Confusion Matrix

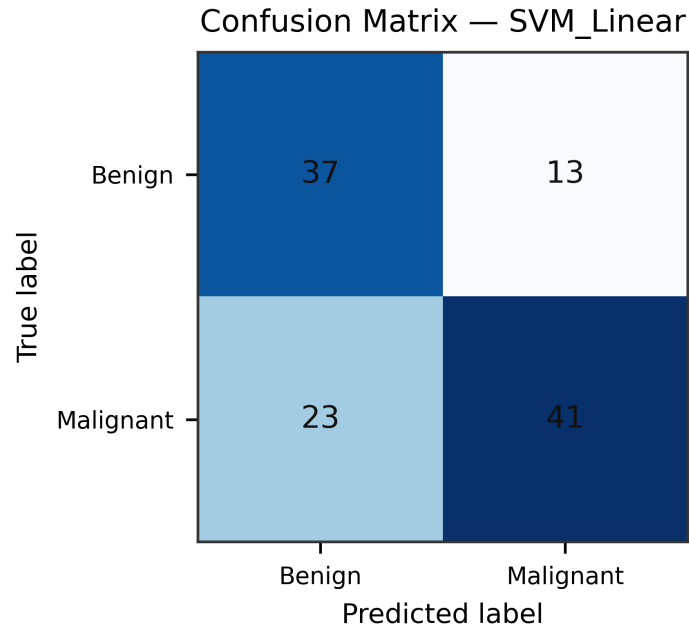


Figure 6: Confusion matrix of the baseline Linear SVM model.

The Linear SVM correctly classified 37 benign and 41 malignant ROIs. It incorrectly classified 13 benign ROIs as malignant and missed 23 malignant ROIs.

The model had higher specificity than sensitivity:

$$Sensitivity \approx 0.64, \quad Specificity \approx 0.74.$$

3.3 Metric Profile

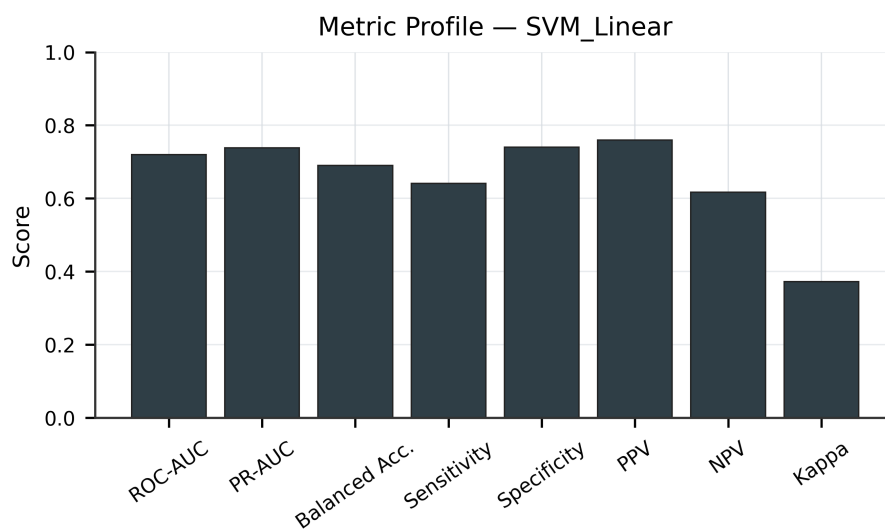


Figure 7: Performance profile of the baseline Linear SVM model.

The model achieved approximately:

- ROC-AUC: 0.72;

- PR-AUC: 0.74;
- balanced accuracy: 0.69;
- sensitivity: 0.64;
- specificity: 0.74;
- positive predictive value: 0.76;
- negative predictive value: 0.62;
- Cohen's kappa: 0.37.

3.4 Feature-Selection Stability

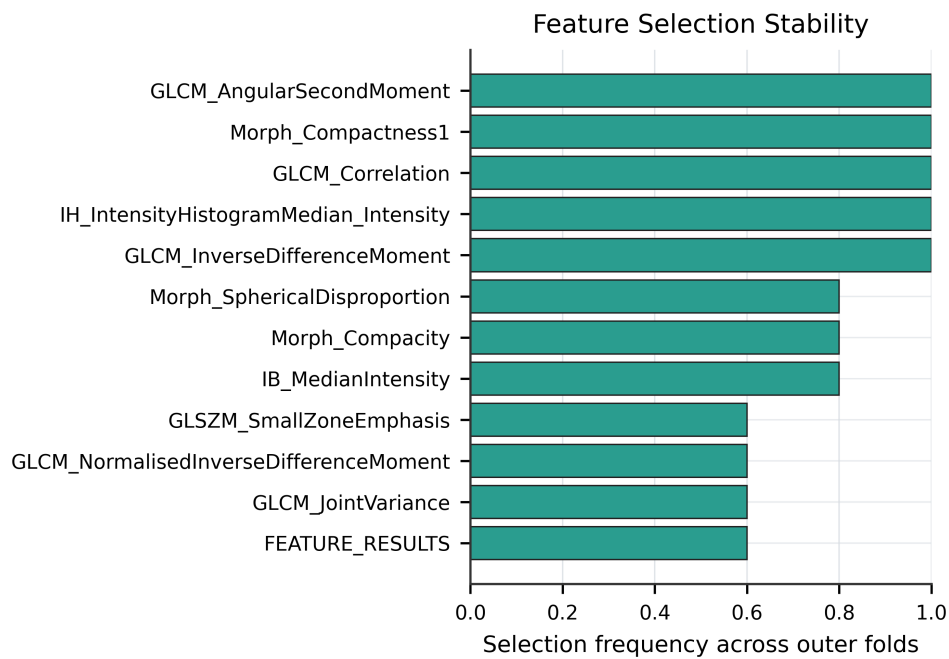


Figure 8: Selection frequency of the most stable baseline features across the five outer folds.

Five features were selected in all five outer folds:

- GLCM_AngularSecondMoment;
- Morph_Compactness1;
- GLCM_Correlation;
- IH_IntensityHistogramMedian_Intensity;
- GLCM_InverseDifferenceMoment.

Several morphological and intensity features were also selected in four of the five folds. This suggests that the most stable predictive information was mainly related to texture, lesion shape, and intensity distribution.

4 Main Experimental Results

4.1 Fold-Level Performance

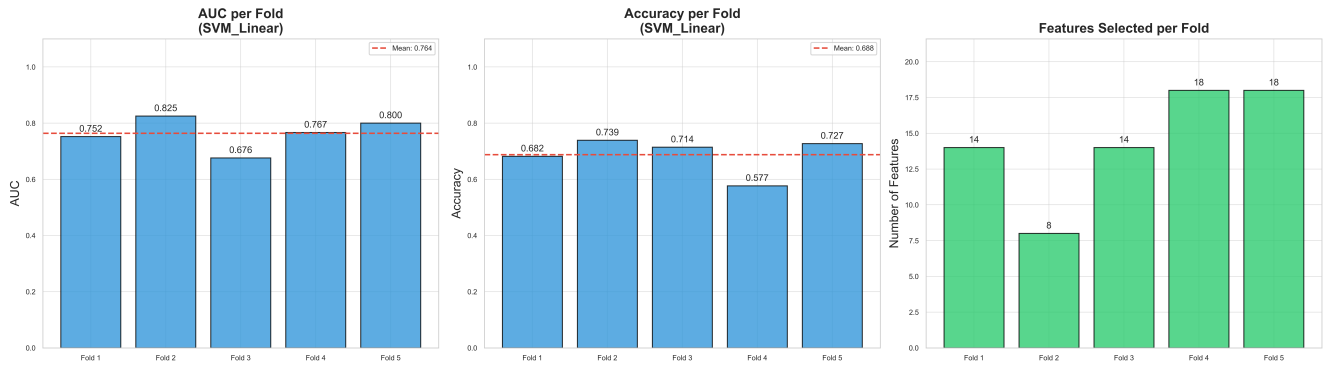


Figure 9: AUC, accuracy, and number of selected features across the five outer folds for the Linear SVM model.

The Linear SVM achieved a mean AUC of 0.764 and a mean accuracy of 0.688 across the five outer folds.

The AUC values ranged from 0.676 to 0.825. Fold 2 showed the best discrimination, while Fold 3 had the lowest AUC. Overall, the AUC remained above 0.67 in all folds, which indicates reasonably stable ranking performance.

Accuracy varied more than AUC, especially in Fold 4, where accuracy decreased to 0.577 despite an AUC of 0.767. This difference shows that the model ranked the samples reasonably well, but the default classification threshold did not work equally well in every fold.

The number of selected features was not fixed. It changed across folds:

14, 8, 14, 18, 18.

This confirms that the Auto-k procedure selected the feature count separately for each training fold. The changes in feature number did not show a simple relationship with performance. For example, Fold 2 achieved the highest AUC using only 8 features, while Fold 4 used 18 features but had the lowest accuracy.

4.2 Selected Run for Detailed Analysis

Logistic Regression without SMOTE had the highest mean AUC. However, the detailed analysis was performed on the ElasticNet-mRMR pipeline with SMOTE because Linear SVM gave a better overall balance across several metrics.

4.3 Uncertainty of the Result

The pooled AUC of Linear SVM was 0.720, with a 95% confidence interval from 0.619 to 0.814.

This interval is relatively wide, which shows that the exact performance is still uncertain.

4.4 Statistical Comparison

The overall comparison between the models was not statistically significant:

$$p = 0.263.$$

5 Analysis and Model Interpretation

Older experiments based on LASSO, RFE, custom mRMR, and hierarchical feature selection were skipped and did not affect the current ADC results.

5.1 SHAP Feature Importance

SHAP analysis was used to understand which features had the largest influence on the selected Linear SVM model.

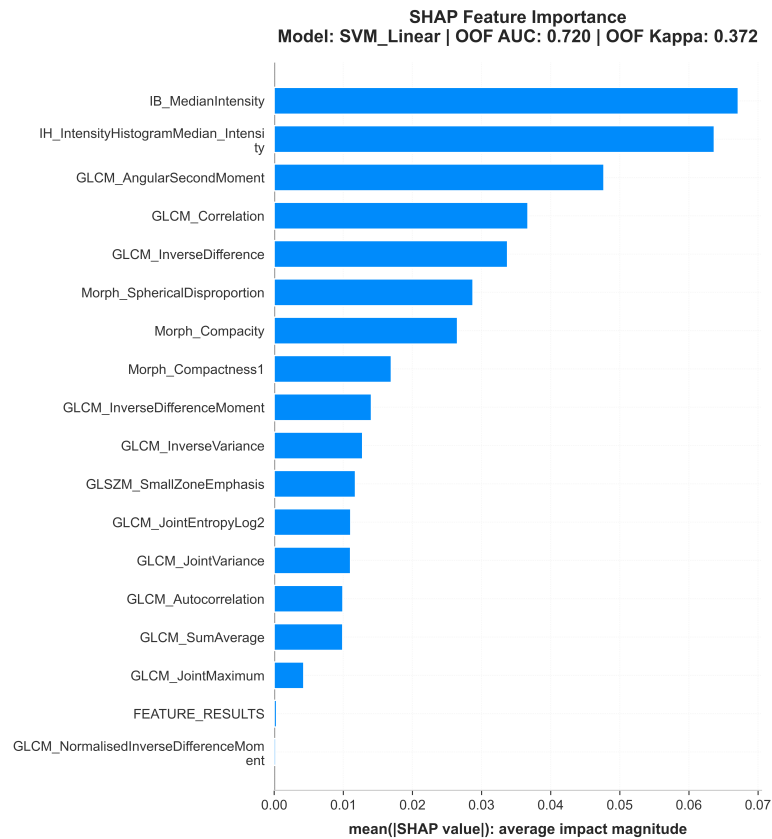


Figure 10: Mean absolute SHAP importance for the selected Linear SVM model.

The most important features were:

- IB_MedianIntensity;
- IH_IntensityHistogramMedian_Intensity;
- GLCM_AngularSecondMoment;
- GLCM_Correlation;
- GLCM_InverseDifference;
- Morph_SphericalDisproportion;
- Morph_Compacity.

The two median-intensity features had the highest average effect. Texture features from GLCM and morphological features also had important roles.

Overall, the model mainly used three types of information:

- signal intensity;
- texture and local spatial patterns;
- lesion shape and compactness.

5.2 Direction of Feature Effects

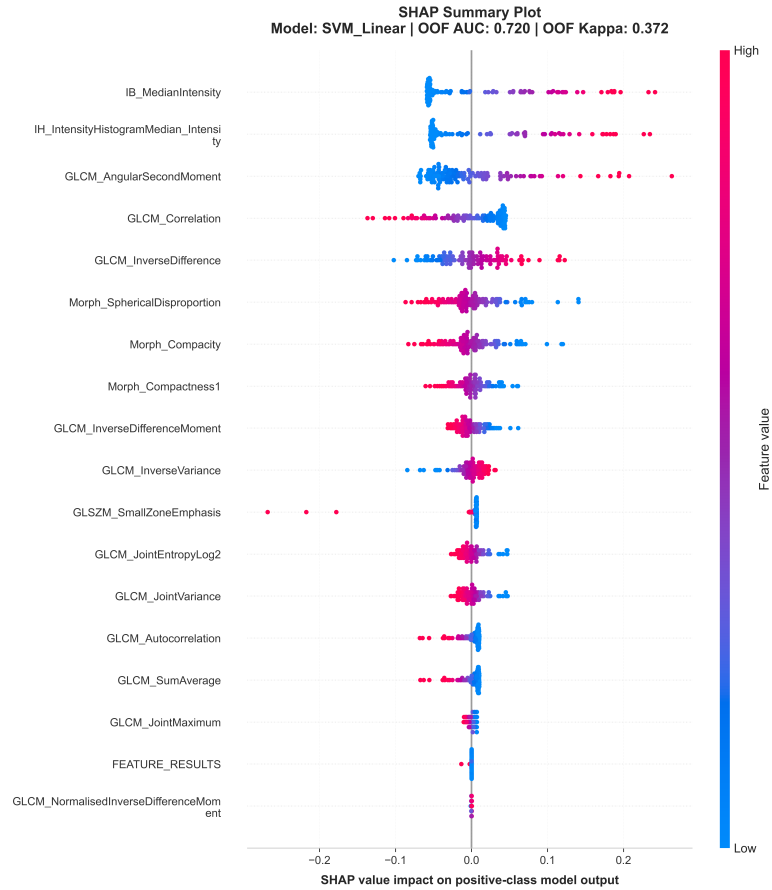


Figure 11: SHAP summary plot showing the direction and size of feature effects on the malignant-class output.

The SHAP summary plot showed both the importance and direction of each feature.

Higher values of the median-intensity features and **GLCM_AngularSecondMoment** generally increased the model output for the malignant class.

Higher values of **GLCM_Correlation**, **Morph_SphericalDisproportion**, and **Morph_Compacity** generally reduced the malignant-class output.

5.3 Feature-Family Contribution and Selection Stability

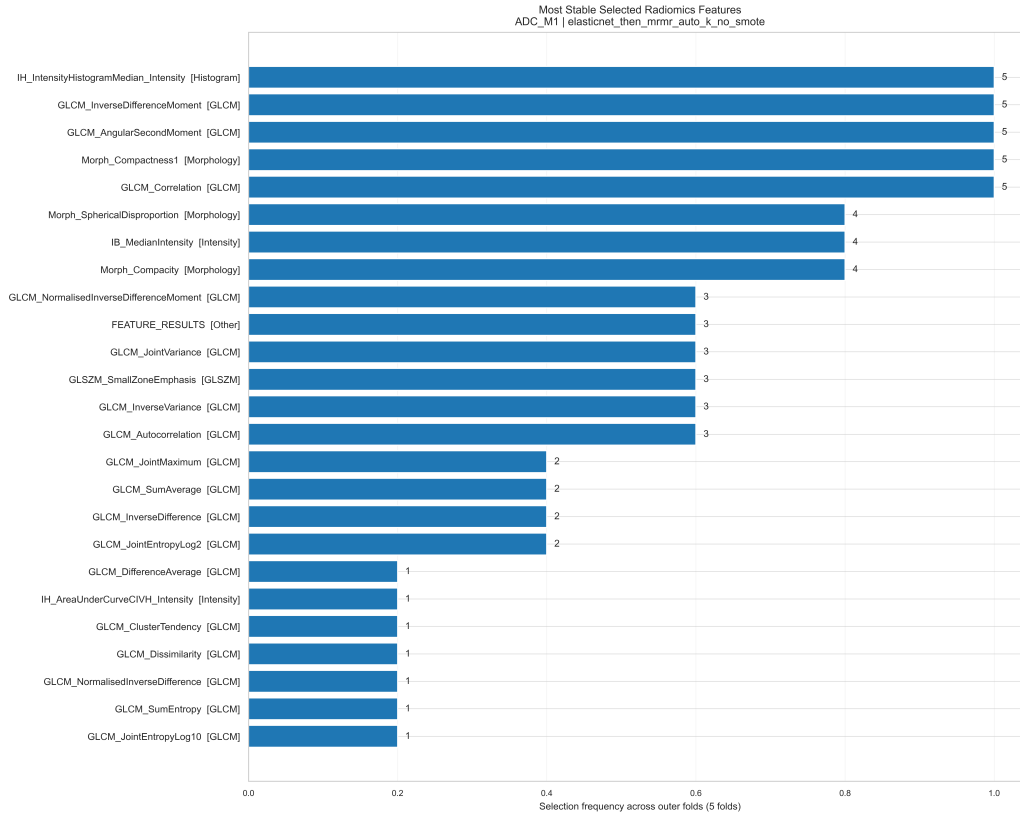


Figure 12: Selection frequency of ADC radiomics features across the five outer folds.

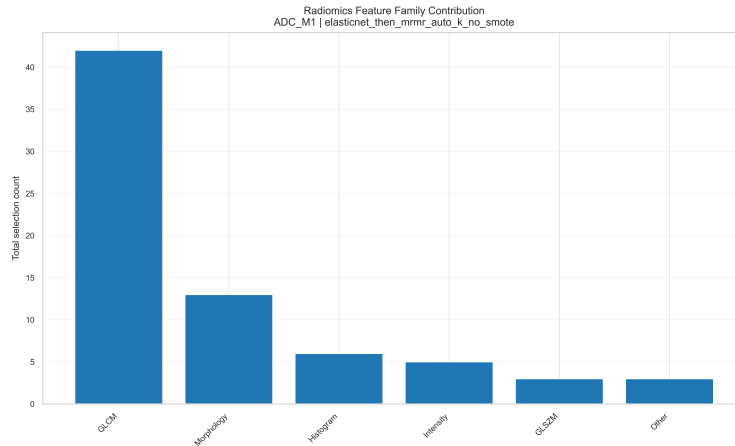


Figure 13: Total number of selected features from each radiomics feature family.

This suggests that the model mainly used texture information, especially spatial grey-level relationships, together with a smaller contribution from lesion shape and intensity.

6 Pairwise Analysis: Post2 and ADC

In the second part of the project, I combined the radiomics features from the Post2 and ADC datasets. The same patients and ROIs were used, but each ROI now had features from both datasets.

The same patient-aware nested cross-validation pipeline was used. Therefore, the main difference from the previous analysis was the larger feature set.

6.1 Combined Dataset

The combined dataset contained:

- 114 ROIs;
- 48 patients;
- 289 radiomics features;
- 50 benign and 64 malignant ROIs.

The number of samples did not increase. Only the number of features increased from 145 in ADC alone to 289 after combining Post2 and ADC.

6.2 Baseline Result

The first baseline used 14 fixed features, mRMR followed by ElasticNet, and no SMOTE.

The best baseline model was Logistic Regression:

$$MeanAUC = 0.801 \pm 0.072,$$

$$PooledAUC = 0.749.$$

6.3 Number of Selected Features

In the Auto-k experiment, the selected numbers of features were:

$$14, 8, 16, 18, 8.$$

The average was 12.8 features.

6.4 Feature Stability

A total of 47 different features were selected across the five folds.

No feature was selected in all five folds. Only one feature appeared in four folds.

This shows that the selected feature set was less stable than in the ADC-only analysis. However, useful features came from both Post2 and ADC.

The most repeated feature was:

$$\text{Post2_IH_IntensityHistogramMode_Intensity.}$$

6.5 Comparison with ADC Alone

The best ADC-only result was:

$$MeanAUC = 0.771 \pm 0.038, \quad PooledAUC = 0.710.$$

The Pairwise baseline result was:

$$MeanAUC = 0.801 \pm 0.072, \quad PooledAUC = 0.749.$$

Therefore, combining Post2 and ADC improved the observed AUC.

However, the Pairwise result had a larger standard deviation and less stable selected features. This means that performance improved, but the result was more sensitive to the patient folds.

7 Triple-Dataset Analysis

In the next stage, I combined the radiomics features from three datasets. Ten different triple combinations were created and evaluated using the same pipeline.

Each combined dataset contained 435 radiomics features. Depending on the available matched ROIs, the number of samples ranged from 113 to 117.

7.1 Best Triple Combinations

The best result was obtained using:

$$\text{ADC} + \text{Post1} + \text{Pre}$$

The selected pipeline was:

$$\text{ElasticNet} + \text{mRMR} + \text{Auto-k} + \text{Linear SVM, without SMOTE.}$$

Its main results were:

$$\text{MeanAUC} = 0.836 \pm 0.088,$$

$$\text{PooledAUC} = 0.799, \quad \text{PooledPR} - \text{AUC} = 0.832.$$

The second-best combination was:

$$\text{ADC} + \text{Post2} + \text{T2}$$

with:

$$\text{MeanAUC} = 0.834 \pm 0.045, \quad \text{PooledAUC} = 0.756.$$

The mean AUC values of the two best combinations were almost equal. However, ADC–Post2–T2 had a smaller standard deviation, which means that its performance was more stable across the folds.

Triple combination	Best model	Mean AUC	SD	Pooled AUC
ADC–Post1–Pre	Linear SVM	0.836	0.088	0.799
ADC–Post2–T2	Linear SVM	0.834	0.045	0.756
ADC–Post1–Post2	Linear SVM	0.795	0.080	0.759
ADC–Post2–Pre	Logistic Regression	0.790	0.052	0.717
ADC–Pre–T2	XGBoost	0.762	0.115	0.666

Table 2: Five highest-performing triple-dataset combinations.

7.2 Fold-Level Performance

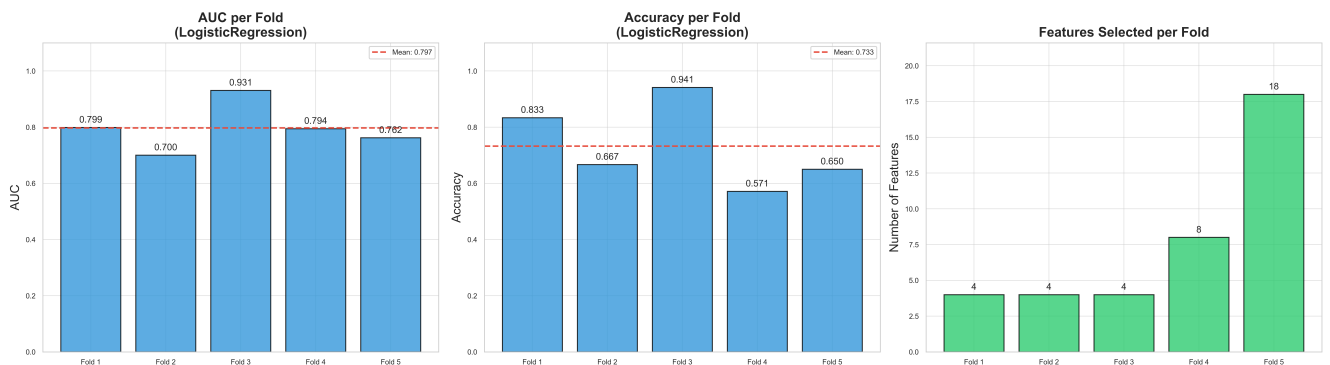


Figure 14: AUC, accuracy, and number of selected features across the five outer folds for the LogisticRegression model.

7.3 Importance of ADC

The five highest-ranked triple combinations all included ADC.

7.4 Feature Selection and Model Type

ElasticNet followed by mRMR was the best feature-selection method in five of the ten triple combinations.

mRMR followed by ElasticNet was best in four combinations, while correlation filtering followed by mRMR was best in only one combination.

Linear SVM was the best classifier in four of the ten combinations.

7.5 Comparison with Earlier Results

The best ADC-only result was:

$$MeanAUC = 0.771.$$

The best Post2–ADC pairwise result was approximately:

$$MeanAUC = 0.801.$$

The best triple result increased to:

$$MeanAUC = 0.836.$$

8 Four-Dataset Results: ADC–Post1–Post2–Pre

This section is the results of the combined **ADC–Post1–Post2–Pre** dataset. The same pipeline was used.

8.1 Best Result

The highest mean AUC was obtained using:

ElasticNet + mRMR + Auto-k + SVM-RBF, without SMOTE.

The main results were:

$$MeanAUC = 0.813 \pm 0.035,$$

$$PooledAUC = 0.765, \quad PooledPR - AUC = 0.824.$$

The small standard deviation shows that the model had relatively stable performance across the five folds.

8.2 Effect of SMOTE

Using SMOTE slightly reduced the mean AUC:

$$0.813 \rightarrow 0.806.$$

However, it improved some other measures, including pooled AUC, balanced accuracy, specificity, and F1-score. Therefore:

- without SMOTE gave the highest mean AUC;
- with SMOTE gave a better balance between benign and malignant classification.

8.3 Selected Features

The best method selected between 4 and 12 features in each fold, with an average of 7.6 features.

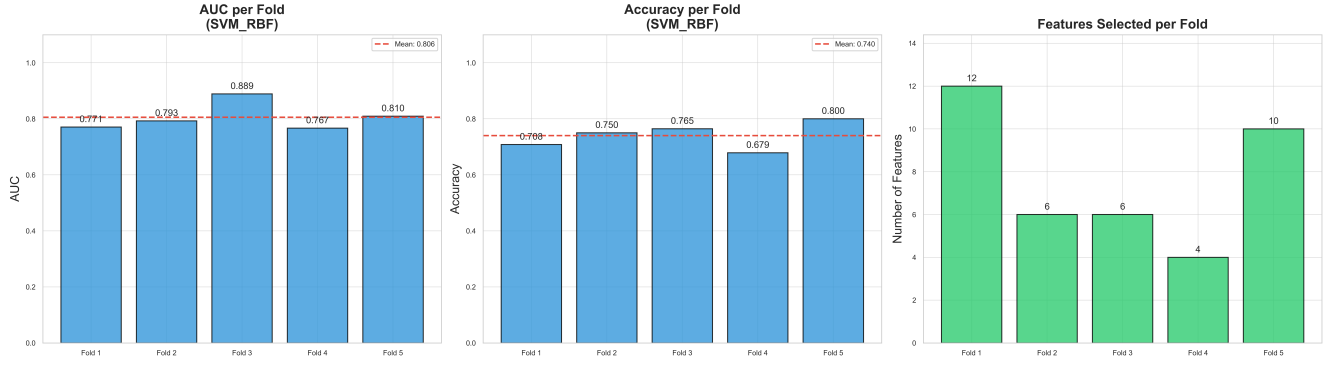


Figure 15: AUC, accuracy, and number of selected features across the five outer folds for the SVM-RBF model.

9 Five-Dataset Results

In the final experiment, all five datasets were combined:

$$\text{Pre} + \text{Post1} + \text{Post2} + \text{ADC} + \text{T2}$$

The combined dataset contained 113 samples and 725 radiomics features.

The best model was:

$$\text{mRMR} + \text{ElasticNet} + \text{Auto-k} + \text{SMOTE} + \text{Logistic Regression}.$$

Its results were:

$$\text{MeanAUC} = 0.733 \pm 0.038, \quad \text{PooledAUC} = 0.718.$$

The model selected about 10 features in each fold.

The result was lower than the best triple- and four-dataset combinations. Therefore, combining all five datasets did not improve performance.